

Christian-Albrechts-Universität zu Kiel
 Institut für Informatik
 Marine Data Science
 Kiel, Germany



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Academic Appointments

Christian-Albrechts-Universität zu Kiel <i>KiTE Research Fellow</i>	Kiel, Germany	2024 – Now
Ukrainian Catholic University <i>Assistant Professor</i>	Lviv, Ukraine	2023 – 2024
Czech Inst. of Informatics, Robotics & Cybernetics <i>Research Fellow</i>	Prague, Czechia	2022 – 2023
Chalmers University of Technology <i>Post-Doctoral Research Scientist</i>	Gothenburg, Sweden	2021
Facebook Reality Labs, AR/VR <i>Post-Doctoral Research Scientist</i>	Pittsburgh, PA	2019 – 2021

Education

Czech Technical University <i>Ph.D., Computer Science</i> Thesis: “Methods for the Rectification of Imaged Coplanar Repeated Patterns”	Prague, Czechia	2020
Czech Technical University <i>M.Sc., Computer Science</i>	Prague, Czechia	2013
The University of North Texas <i>B.Sc., Mathematics</i>	Denton, TX	2002

Publications

- J. Pritts** and K. Köser. Robust Local Optimization Done Right. In *ACCV*, 2026
- J. Pritts**, F. Seegräber, and K. Köser. RANSAC Scoring Done Right. In *NeurIPS*, 2026
- J. Pritts**, T. Sittart, H. Sauer, S. Janßen, F. Seegräber, D. Nakath, and K. Köser. BlobBoards: Robust Markers for Accurate Pose. In *BMVC*, 2026
- E. Dexheimer, P. Peluse, J. Chen, **J. Pritts**, and M. Kaess. Information-theoretic online multi-camera extrinsic calibration. *IEEE Robotics and Automation Letters*, 7(2):4757–4764, 2022
- Y. Lochman, K. Liepeshov, J. Chen, M. Perdoch, C. Zach, and **J. Pritts**. Babelcalib: a universal approach to calibrating central cameras. In *ICCV*, 2021
- Y. Lochman, O. Dobosevych, R. Hryniv, and **J. Pritts**. Minimal Solvers for Single-View Lens-Distorted Camera Auto-Calibration. In *WACV*, 2021
- J. Pritts**, Z. Kukelova, V. Larsson, Y. Lochman, and O. Chum. Minimal solvers for rectifying from radially-distorted conjugate translations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(11):3931–3948, 2021
- J. Pritts**, Z. Kukelova, V. Larsson, Y. Lochman, and O. Chum. Minimal Solvers for Rectifying from Radially-Distorted Scales and Change of Scales. *International Journal of Computer Vision*, 128(4):950–968, 2020
- J. Pritts**, Z. Kukelova, V. Larsson, and O. Chum. Rectification from Radially-Distorted Scales. In *ACCV*, 2018
- J. Pritts**, Z. Kukelova, V. Larsson, and O. Chum. Radially-Distorted Conjugate Translations. In *CVPR*, 2018

J. Pritts, D. Rozumnyi, M. P. Kumar, and O. Chum. Coplanar Repeats by Energy Minimization. In *BMVC*, 2016

J. Pritts, O. Chum, and J. Matas. Detection, Rectification and Segmentation of Coplanar Repeated Patterns. In *CVPR*, 2014

J. Pritts, O. Chum, and J. Matas. Approximate Models for Fast and Accurate Epipolar Geometry Estimation. In *IVCNZ*, 2013

Recognition and Awards

Asian Conference on Computer Vision (ACCV) Saburo Tsuji Best Paper Award
for “Rectification from Radially-Distorted Scales” 2018

Image and Vision Computing New Zealand (IVCVNZ) Best Paper Award
for “Approximate Models for Fast and Accurate Epipolar Geometry Estimation” 2013

Professional Association Memberships

ELLIS Scholar 2024 – Now

Kiel Marine Science Associate 2025 – Now

Funding

Principal Kiel Marine Science (KMS) 2025
“Enhancing Underwater Robotics for Visual Mapping in Turbid
Environments”
Awarded: €10,000

Principal Kiel Training for Excellence (KiTE) co-funded by MSCA 2024 – 2027
“Live and Robust Visual Localization and Mapping in Coastal Waters”
Awarded: \$225,000

Contributing Facebook Sponsored Research Agreement 2020 – 2021
“In-the-field Extrinsic Calibration of Multi-camera Systems”
Awarded: \$150,000

Principal Facebook Sponsored Research Agreement 2020
“Calibration of Head-Mounted Multi-Camera Capture Systems”
Awarded: \$25,000

Professional Service Activities

Reviewer CVPR, ICCV, ECCV, 3DV, WACV, ACCV, ICRA, IJCV,
PAMI

Conference Organizer Poster Chair 3DV 2022

Tutorial Organizer “Affine Correspondences and Their Applications” 3DV 2022

Tutorial Organizer “Affine Correspondences and Their Applications” CVPR 2022

Teaching

3D Computer Vision 2025 – 2026
Substitute Instructor, M.Sc. level, Kiel University

Image Retrieval 2017 – 2018
Instructor, M.Sc. level, Ukrainian Catholic University

Pattern Recognition and Machine Learning 2013 – 2016
Teaching Assistant, B.Sc. level, Czech Technical University in Prague

Advising

M.Sc. Students

Ties Harder	Kiel University Thesis: “Boundary Movement Aware Differentiable Rendering for Visual Calibration Tasks”	in progress
Philipp Böll	Kiel University Thesis: “Underwater Light Rig Auto-Calibration from Asynchronous Cameras”	in progress
Hendrik Sauer	Kiel University Thesis: “Affine-Equivariant Neural Networks for the Feature Matching Problem”	2026
Till Sittart	Kiel University Thesis: “Affine-Equivariant Blob Targets for Robust Absolute Pose in Robotics”	2026
Liam Boddin	Kiel University Thesis: “Underwater Calibration of Near-point Anisotropic Lights”	2025
Igor Babin	Ukrainian Catholic University Thesis: “Image Inpainting in Latent Space”	2024
Dmytro Nadobko	Ukrainian Catholic University Thesis: “Supervised Learning of Correspondence Volumes for Coplanar Repetitive Patterns”	2023
Andrii Stadnik	Ukrainian Catholic University Thesis: “Corner Localization and Camera Calibration from Imaged Lattices”	2023
Yaroslava Lochman	Ukrainian Catholic University Thesis: “Minimal Solvers for Single-View Auto-Calibration”	2020

B.Sc. Students

Yaroslav Romanus	Ukrainian Catholic University Thesis: “Applying Motion in Latent Space”	2024
Ostap Viniavskyi	Ukrainian Catholic University Thesis: “Learning Discriminative Context-Aware Keypoints Representations for Resolving Ambiguous Matches”	2021
Kostiantyn Liepieshov	Ukrainian Catholic University Thesis: “Manhattan-Frame Detection in Lens-Distorted Images”	2021

Invited Talks and Lectures

Fishbowl Discussion, Annual Kiel Marine Science Retreat “NextGen Perspectives”	September 2026 Kiel, Germany
3DV Tutorial Talk “Just One Image is All It Takes...”	September 12, 2022 Prague, Czechia
CVPR Tutorial Talk “Just One Image is All It Takes...”	June 19, 2022 New Orleans, LA
Facebook Reality Labs “Minimal Solvers and Multi-Model Estimators for Image Undistortion and Scene-Plane Rectification”	June 14, 2018 Pittsburgh, PA
The Aspen Institute’s 2018 Young Leader’s Program “Opportunities and Risks of Artificial Intelligence”	March 17, 2018 Tále, Slovakia
The Eastern European Computer Vision Conference “Detection, Rectification, and Segmentation of Coplanar Repeated Patterns”	July 14, 2017 Odesa, Ukraine
The 34th Pattern Recognition and Computer Vision Colloquium “Detection, Rectification, and Segmentation of Coplanar Repeated Patterns”	April 3, 2014 Prague, Czechia